

Review

Trends and Applications of Artificial Intelligence in Project Management

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Abstract: The integration of artificial intelligence (AI) into project management (PM) transforms how projects are planned, executed, and monitored. The main objective of this study is to provide a comprehensive bibliometric analysis exploring trends, thematic areas, and future directions in AI applications in project management by examining publications from the last decade. This research uncovers dominant themes such as machine learning, decision making, information management, and resource optimization. The findings highlight the growing use of AI to enhance efficiency, accuracy, and innovation in PM processes, with recent trends favoring data-driven approaches and emerging technologies like generative AI. Geographically, China, India, and the United States lead in publications, while the United Kingdom and Australia show a high citation impact. The research landscape, including AI-enhanced decision-making frameworks and cost analysis, demonstrates the diversity of AI applications in PM. An increased interest in the use of generative AI and its impact on PM and project managers was observed. This analysis contributes to the field by offering a structured overview of research trends, defining the challenges and opportunities for integrating AI into project management practices and offering perspectives on emerging technologies.



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1. Introduction

Artificial intelligence (AI) has emerged as a transformative force in various industries and in project management (PM). The integration of AI technologies into PM practices has the potential to revolutionize how projects are planned, executed, and monitored [1,2]. The need for data-driven decision-making processes, more effective resource allocation, and the growing complexity of projects are the main forces behind this paradigm change, where most projects are immersed in industry 4.0 [3–6].

The application of AI in PM comprehends a wide range of techniques and methodologies, including machine learning, natural language processing, and predictive analytics. These technologies enable project managers to enhance their capabilities in areas such as risk assessment, resource optimization, and performance forecasting [7–10]. Organizations may be able to increase overall efficiency, lower expenses, and raise project success rates by utilizing AI [11,12].

The complex effects of AI on PM techniques have been brought to light by recent studies. For example, AI-powered tools can analyze vast amounts of historical project data

to identify patterns and trends, thereby assisting in more accurate project planning and estimation [13]. Furthermore, AI algorithms can continuously monitor project progress, detecting potential issues before they escalate and suggesting corrective actions [14,15]. There are many advantages to integrating AI into PM, such as the automation of tedious tasks, enhanced data analysis capabilities, and enhanced team member engagement and communication. Additionally, AI can improve risk management, optimize resource allocation, and produce more precise project outcome estimates [16]. These advantages contribute to increased efficiency, reduced human error, and more informed decision making throughout the project lifecycle. However, the integration of AI into PM is not without challenges. Concerns regarding data privacy, the need for specialized skills, and the potential for over-reliance on automated systems have been raised by researchers and practitioners alike [17–19]. In order to optimize the advantages while minimizing any possible threats, a well-rounded strategy that blends AI skills with human knowledge is frequently recommended.

The rapid advancement of AI technologies has led to the development of various machine learning architectures, each with its own strengths and applications in PM. The most important models being explored for PM tasks are the following (Figure 1): (i) Long Short-Term Memory (LSTM) networks are a type of Recurrent Neural Network and are particularly adept at handling sequential data and long-term dependencies [20]; (ii) Convolutional Neural Networks (CNNs), traditionally used in image processing, have found applications in PM through their ability to identify spatial patterns in data [21]; (iii) Gated Recurrent Units (GRUs), similar to LSTMs but with a simpler structure, offer a balance between computational efficiency and performance [22]; (iv) Multilayer Perceptrons (MLPs), a class of feedforward artificial neural networks, are versatile and can be applied to a wide range of PM tasks [23]; and (v) Recurrent Neural Networks (RNNs), the foundation for more complex architectures like LSTMs and GRUs, are particularly useful for processing sequential data [24]. These models have demonstrated promise in forecasting important project factors such as expected costs, staff requirements, and completion time.

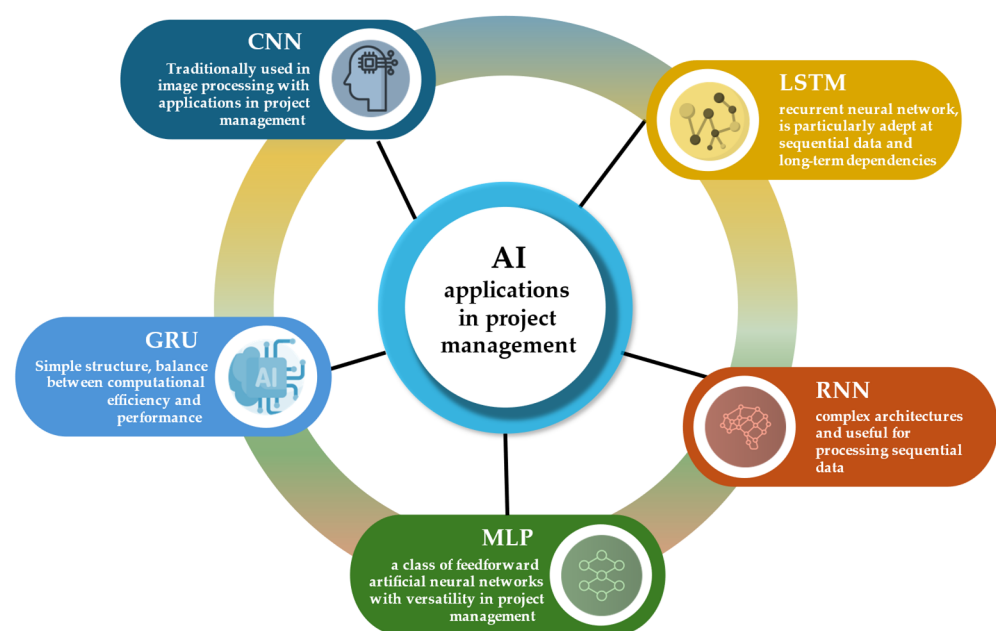


Figure 1. AI applications in project management.

Comparative analyses of these distinct models are becoming more and more necessary as the field of AI in PM develops in order to ascertain their efficacy in diverse project

settings [25–27]. These assessments can offer important information about which AI methods are most appropriate for PM responsibilities and how to use them effectively to improve project results. In this regard, while the state of the art includes several systematic reviews addressing the role of AI in project performance [28], there is a clear need for a bibliometric analysis to provide a comprehensive overview of trends and research dynamics in this area.

This paper offers a comprehensive bibliometric analysis of AI in project management, examining key trends, thematic areas, and future prospects. Moreover, this study explores the growth of AI in project management, highlighting emerging technologies, publication patterns across different regions, and citation impact. It also assesses the extent of international collaboration in the field. Finally, this paper will address the benefits, challenges, and potential future directions for the application of AI in project management.

2. Materials and Methods

As this study aimed at analyzing the role of AI in the context of PM, a bibliometric analysis approach was adopted since it is deemed suitable for the examination of broad topics [29–31]. On the one hand, to identify relevant studies, detailed selection criteria were established, focusing on the specific use of artificial intelligence in project management. On the other hand, to analyze the related data derived from Scopus and Web of Science, the Bibliometrix tool was used [32]. These databases were selected due to their high relevance, rigorous indexing criteria, and widespread use in bibliometric studies. Both Scopus and Web of Science provide comprehensive coverage of high-impact scientific literature, ensuring that the dataset used for analysis is robust, reliable, and representative of the research field [33]. Moreover, the data generated are usable by the Bibliometrix tool.

Additionally, to report the data collection process, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [34] were used. Given the general scope of this study, a more generic query was used: (“artificial intelligence” or “ai”) and (“project management”). The specific query consisted of keywords that are highly relevant to the topic, such as artificial intelligence, AI, and PM. This allowed for a transparent and replicable data extraction method.

The results were last updated through the query used in January 2025. The restrictions set were for the documents to have been published in English and in the period of 2015–2024. This study initially considered publications from 2015 to 2024; however, after applying the selection criteria, it was observed that few relevant studies on AI applications in project management were published between 2015 and 2018. As a result, while the study period remains 2015–2024, most analyzed publications originate from 2019 onwards. As explained in Figure 2, 410 documents were identified from Scopus and 184 from Web of Science. Of the total 594 documents, 148 were duplicates and were, thus, removed. Hence, 446 documents were assessed for eligibility. The inclusion criterion set was for the study to focus primarily on the use of artificial intelligence in the context of PM. To check whether a document met the inclusion criterion specified, its title and abstract were manually assessed. A total of 328 documents were excluded as they did not meet the inclusion criteria, 2 documents were removed as they were editorials, and 1 document was removed since it was a book. As a result, a total of 115 documents remained and are examined in this study.

Regarding data analysis techniques, the Bibliometrix tool was applied to perform a comprehensive bibliometric analysis, which included citation analysis, co-citation analysis, cluster analysis, and keyword co-occurrence analysis. In addition to the analysis of the document collection through Bibliometrix, a descriptive analysis of its main characteristics was also carried out. Emphasis was placed on the analysis of the annual number of publications and citations, on identifying the main sources, and examining the authors’

countries. This facilitated an in-depth understanding of trends, patterns, and relationships in the research field.

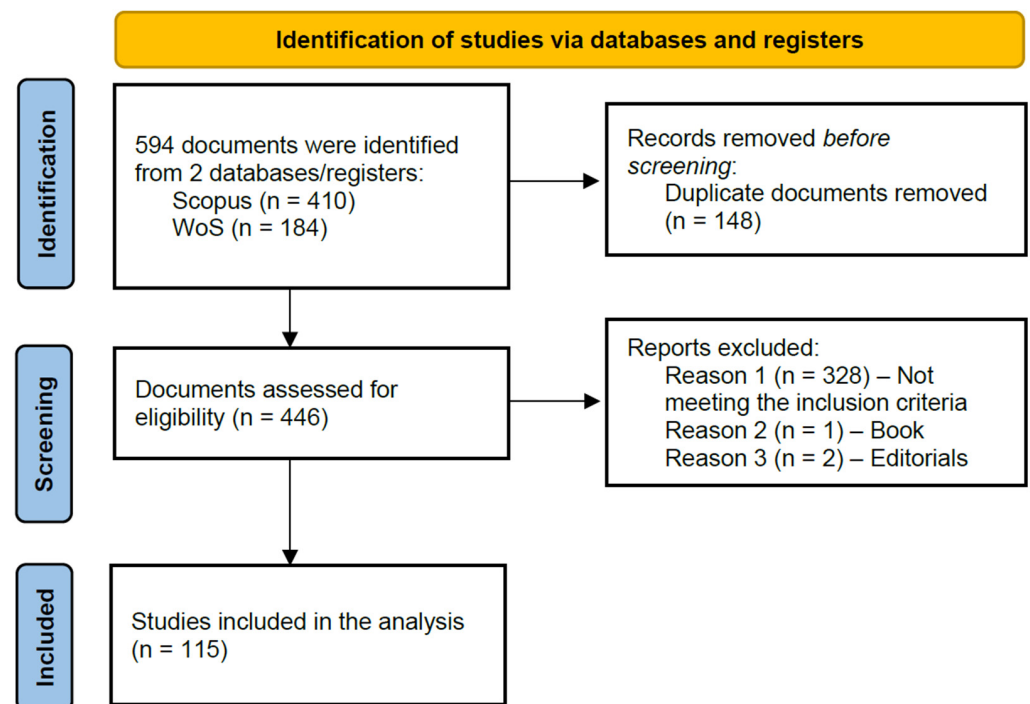


Figure 2. Identification of studies via PRISMA 2020 protocol.

3. Results

As can be seen in Figure 3, although the search for relevant documents was focused on documents that were published during 2015–2024, documents within the scope of this study were noticed starting from 2019. Therefore, while this study covers the entire period (2015–2024), the bibliometric analysis is effectively focused on 2019–2024 due to the low number of relevant publications in the earlier years. Hence, the document collection had 115 documents which were published from 2019 to 2024. The documents were published in 98 sources, by 302 authors from 44 different countries. Among the documents, 21 (18.3%) were single-authored. On average, the documents had an age of 2.3 years and an annual growth rate of 70.32% was noticed. The vast majority of documents were published as journal articles or conference papers. The documents had on average 2.92 co-authors and received 6.7 citations. The international co-authorships accounted for 11.3%.

Most documents were published in 2024 (37.4%), followed by 2023, in which 34 documents (29.6%) were published (Figure 4). Given that the publication of the documents spans across six years (2019–2024), the time period was too short to define distinct time periods regarding the maturation of the topic. Hence, based solely on the number of published documents, three periods are set and examined in the text further below: (i) 2019–2020; (ii) 2021–2022; and (iii) 2023–2024. Additionally, based on the results presented in Table 1, the seven documents published in 2020 had the highest mean total citations per document ($\text{MeanTCperDoc} = 39.71$) and per year ($\text{MeanTCperYear} = 6.62$). However, due to the citable years of the documents and the fact that most were published in the last two years, it can be inferred that these outcomes might change in the future.

Furthermore, the sources in which the documents were published were also examined through the use of Bradford's law. A total of three clusters were created to categorize the sources based on their relevancy with the first cluster having sources that are regarded as more relevant to the topic. The first cluster consisted of 21 sources in which 38 documents

were published, the second cluster had 40 sources in which 40 documents were published, and the third cluster had 37 sources in which 37 documents were published. The related outcomes are presented in Table 2. Having each published three related documents, “Applied Sciences”, “ITNOW”, “Lecture Notes in Networks and Systems”, and “Sustainability” were the top sources in cluster one. In total, nine sources had published two documents, and the remaining eighty-five sources had published a single document relevant to the topic. However, the relatively low number of published documents per source did not allow for a more in-depth analysis of the local impact of the documents within the document collection.

DESCRIPTION	RESULTS
MAIN INFORMATION ABOUT DATA	
Timespan	2019:2024
Sources (Journals, Books, etc.)	98
Documents	115
Annual Growth Rate %	70.32
Document Average Age	2.3
Average citations per doc	6.704
DOCUMENT TYPES	
Article	50
Book chapter	9
Conference/Proceedings paper	50
Review	6
DOCUMENT CONTENTS	
Keywords Plus (ID)	459
Author Keywords (DE)	285
AUTHORS	
Authors	302
Authors of single-authored docs	19
AUTHORS' COLLABORATION	
Single-authored docs	21
Co-authors per doc	2.92
International co-authorships %	11.3

Figure 3. Document collection.

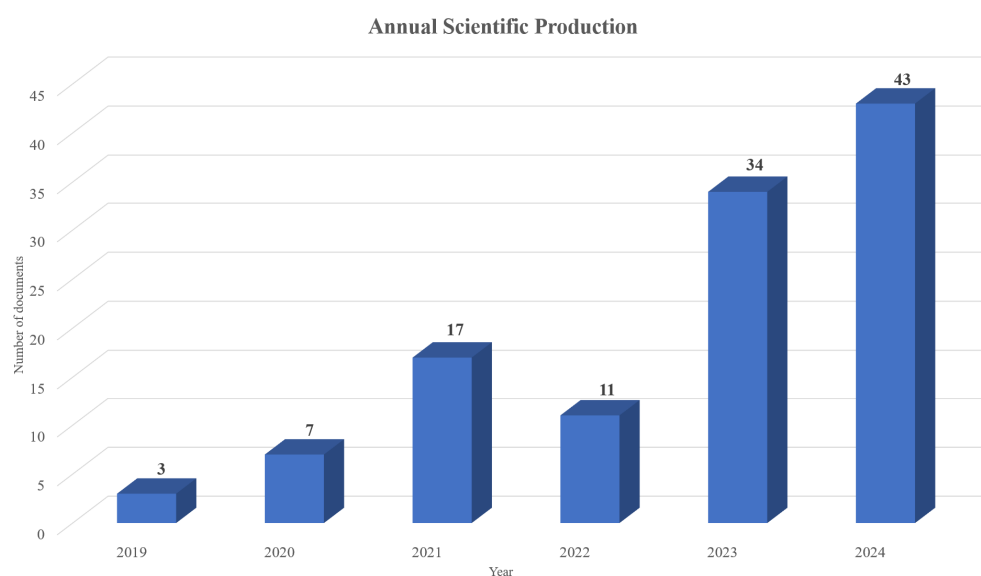


Figure 4. Distribution of annual publications.

Table 1. Document average citations per year.

Year	MeanTCperDoc	N	MeanTCperYear	CitableYears
2019	12.33	3	1.76	7
2020	39.71	7	6.62	6
2021	11.53	17	2.31	5
2022	11.27	11	2.82	4
2023	3.09	34	1.03	3
2024	0.72	43	0.36	2

Table 2. Sources of cluster 1 with at least two relevant documents published.

Journal	Rank	Freq.	cumFreq.	Cluster
<i>Applied Sciences</i>	1	3	3	Cluster 1
<i>ITNOW</i>	2	3	6	Cluster 1
<i>Lecture Notes in Networks and Systems</i>	3	3	9	Cluster 1
<i>Sustainability</i>	4	3	12	Cluster 1
<i>IEEE International Conference on Software Engineering and Artificial Intelligence (SEAI)</i>	5	2	14	Cluster 1
<i>AIP Conference Proceedings</i>	6	2	16	Cluster 1
<i>ASEE Annual Conference and Exposition, Conference Proceedings</i>	7	2	18	Cluster 1
<i>Buildings</i>	8	2	20	Cluster 1
<i>Built Environment Project and Asset Management</i>	9	2	22	Cluster 1
<i>CEUR Workshop Proceedings</i>	10	2	24	Cluster 1
<i>IEEE Engineering Management Review</i>	11	2	26	Cluster 1
<i>International Journal of Advanced Computer Science and Applications</i>	12	2	28	Cluster 1
<i>Project Management Journal</i>	13	2	30	Cluster 1

Moreover, based on Lotka's law, only four authors contributed to three documents (1.3%), two authors contributed twenty-six documents (8.6%), and two hundred and seventy-two authors contributed to one document (90.1%). Given the recency of the topic and the fact that most studies were published in the last two years, it is expected that more authors will be involved in this field of study, and more will contribute to additional studies. Additionally, the authors came from 44 countries when considering the corresponding author's country or the country of the first author. In Table 3, the countries that published at least four documents are presented. China, India, and the United States were the three countries with the most documents. Among all countries, India (SCP = 11) and China (SCP = 10) had the highest intra-country (SCP) collaboration rate while the United Kingdom (MCP = 2) and Australia (MCP2) had the highest inter-country (MCP) collaboration rate.

Table 3. Countries' scientific production.

Country	Articles	SCP	MCP	Freq	MCP_Ratio
China	11	10	1	0.096	0.091
India	11	11	0	0.096	0
United States	9	9	0	0.078	0
United Kingdom	8	6	2	0.07	0.25
Australia	5	3	2	0.043	0.4
Italy	5	5	0	0.043	0
Spain	5	4	1	0.043	0.2
Germany	4	4	0	0.035	0
Ukraine	4	4	0	0.035	0

The authors' affiliations were also examined. When looking at the total citations received per country, the United Kingdom (TC = 151), Vietnam (TC = 70), Australia

(TC = 69), and South Korea (TC = 67) were the countries that received the most citations, as can be seen in Table 4. These high citation impacts may be attributed to the strong emphasis on innovation and the early adoption of AI technologies in PM within these countries. Moreover, given the 11.3% international co-authorship rate observed, Figure 5 presents the collaboration network among them, in which five distinct clusters can be noticed. In this regard, this low international co-authorship rate may be influenced by the technical complexity of AI research. Not all countries have equal access to specialized resources like high-performance computing and private datasets, as well as expertise in data science and machine learning. These restrictions may make it more difficult to collaborate across borders, underscoring the necessity of programs that support multidisciplinary and global research collaborations in this area.

Table 4. Top 10 countries that received the most citations.

Country	TC	Average Article Citations
United Kingdom	151	18.9
Vietnam	70	70
Australia	69	13.8
South Korea	67	33.5
China	47	4.3
Singapore	47	47
United States	47	5.2
Germany	45	11.2
Spain	44	8.8
Iceland	31	10.3

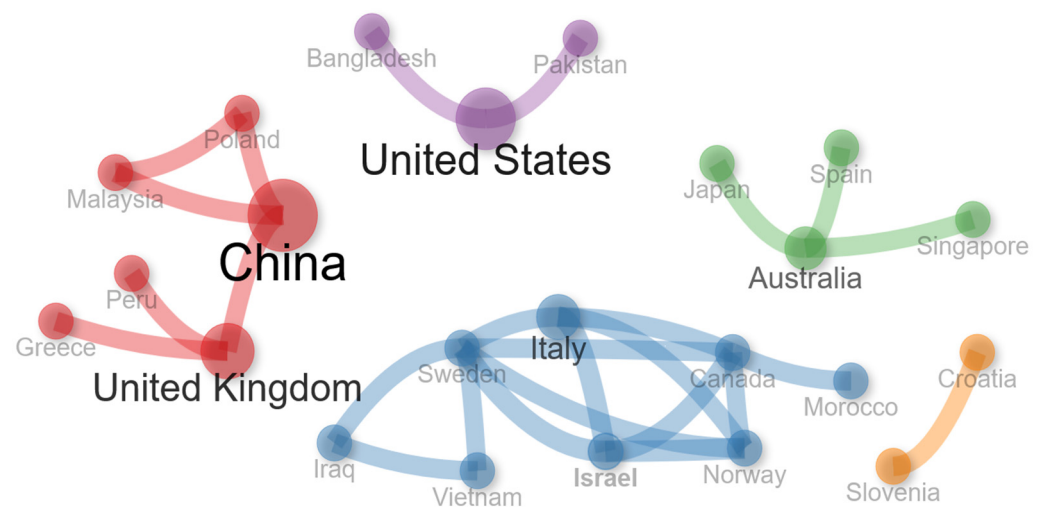


Figure 5. Countries' collaboration.

4. Discussion

This bibliometric research uses both quantitative and graphical information to give an extensive overview of AI trends and applications in PM. The results provide insight into both well-known and new subjects, allowing for a more thorough comprehension of the changing environment in this field.

4.1. Keyword Analysis

Keywords Plus, generated automatically by databases, and Author Keywords, selected by researchers, present valuable understandings into the focus of a field. In this study, “artificial intelligence” and “project management” are the most dominant themes, appearing with high frequencies of 46 and 43 in Keywords Plus and 97 and 46 in Author

Keywords, respectively (Figures 6 and 7). This dominance is largely due to their inclusion in the query used for the bibliometric review, reflecting their foundational role in research exploring AI's integration into PM practices. Moreover, this point of convergence represents a developing trend in which AI-driven solutions are being used to address PM difficulties and enhance decision making, procedures, and results. These results demonstrate how artificial intelligence (AI), a game-changing technology, has emerged as a key component in the development of project management (PM), transforming conventional methods and bringing in new conceptions of effectiveness, agility, and accuracy. Notably, terms such as “machine learning”, “decision making”, and “information management” appear prominently, suggesting a growing emphasis on integrating advanced computational techniques to optimize decision-making processes and enhance information flow in project environments.

Most Common Keywords Plus

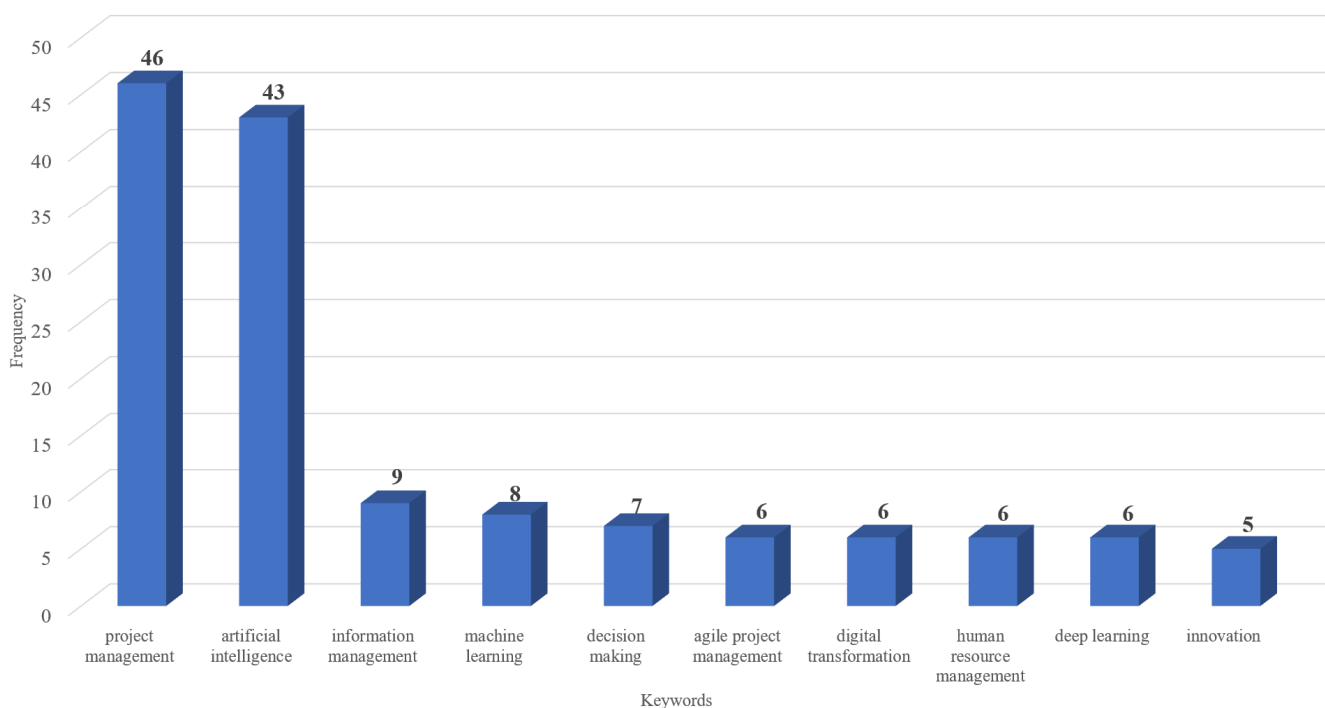


Figure 6. Most common Keywords Plus.

The prominence of “machine learning” highlights its versatility in predicting project outcomes, assessing risks and automating routine tasks [35,36]. Machine learning algorithms are increasingly used for predicting project outcomes by analyzing historical data, enabling the early detection of potential delays, cost overruns, and resource constraints. For example, machine learning has demonstrated significant potential in “smart construction” projects, where it facilitates enhanced automation, safety, and efficiency [17]. Routine processes like risk assessments, resource scheduling, and performance monitoring can be automated with the help of machine learning. Organizations can increase accuracy and production while lessening the cognitive strain on teams and project managers by automating these procedures.

The frequent appearance of “decision making” and “information management” highlights the critical role of data-driven approaches in PM. Large volumes of data are produced by modern projects, encompassing everything from team performance and market trends to financial indicators and resource utilization. Tools with AI capabilities are essential for digesting this data and turning it into insights that can be used. These tools improve the overall quality of information accessible for decision making by minimizing human

mistakes and reducing manual labor. “Information management” reflects the increasing complexity of data in modern projects, where AI-powered tools can process, analyze, and extract actionable insights, significantly reducing manual efforts and errors [37]. Similarly, the frequent appearance of “decision making” as a keyword highlights the role of AI-driven tools in supporting strategic decisions by processing large volumes of data and offering predictive insights, reducing cognitive biases, and enhancing decision accuracy. This capability is particularly relevant in complex and dynamic project environments, where timely and informed decision making is critical [38].

Most Common Author's Keywords

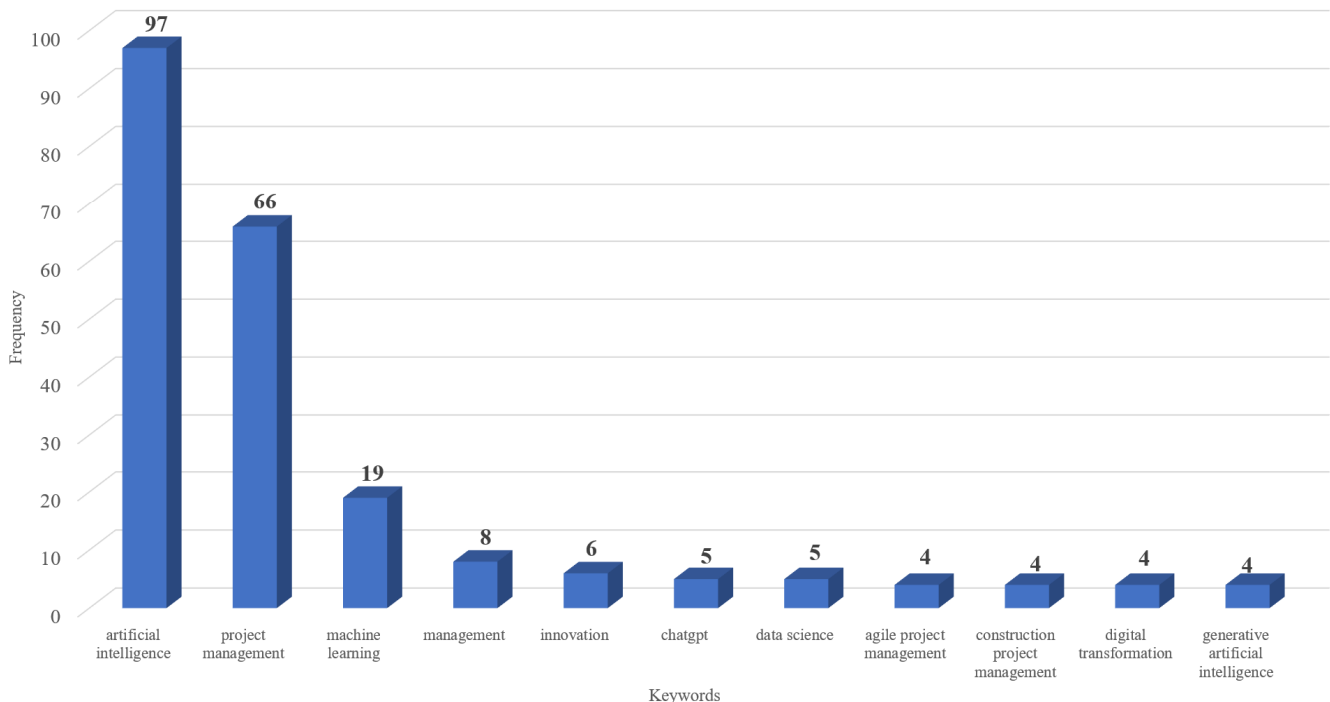


Figure 7. Most common Author Keywords.

The inclusion of “digital transformation” as a keyword underscores the profound impact of AI on digitizing project workflows. Digital transformation initiatives often involve the integration of AI tools to enhance collaboration, streamline processes, and improve transparency. For example, cloud-based platforms powered by AI facilitate real-time updates, enabling project teams to track progress, share insights, and respond to changes with agility [39]. This is especially important for remote teams or international projects where effective coordination and communication are essential. Additionally, the connection to “agile methodologies” emphasizes how AI and iterative project frameworks function together. AI helps agile projects by offering predictive insights, adaptable planning, and real-time analytics that let teams react swiftly to changing needs. Teams can learn from previous iterations and adapt their methods to produce better results thanks to the alignment of AI and agile principles, which promotes a culture of continuous improvement.

Emerging technologies such as “ChatGPT”, “data science”, and “generative AI” are reshaping the situation of PM. ChatGPT, as a conversational AI tool, has gained traction for its ability to facilitate communication, automate report generation, and provide instant access to information. Project managers can use ChatGPT to draft project charters, prepare meeting agendas, and respond to stakeholder inquiries, saving valuable time and ensuring consistency in communication. Real-world examples include its application in construction projects, where ChatGPT has been used to automate client updates and progress reports,

or in software development teams, where it helps manage documentation and client interactions [40,41].

Regarding “data science”, this vast field that serves as the basis for many AI-driven project management systems is also improving the ability to allocate resources optimally, assess project success in real time, and derive actionable insights from complex information. For instance, businesses employ data science methods to find inefficiencies, suggest fixes, and assess how cost-effective various project approaches are. Data science has been used in real-world contexts to optimize team assignments and job distribution in IT projects or to forecast delays in construction projects using previous data [42].

One particularly innovative advancement is generative AI [43], which holds immense potential for automating strategic and creative tasks within PM. Generative AI enables the creation of design concepts, project scenario simulations, and decision support systems that assist teams in exploring new possibilities and driving innovation. In industries such as marketing and construction, generative AI is used to develop customized campaign strategies or optimize building layouts, leading to more effective and creative project outcomes. For instance, in the realm of urban planning, generative AI has been utilized to propose novel city layouts that optimize both space and energy consumption.

The keyword network visualization organizes the interconnected terms into five distinct clusters, each reflecting a unique thematic focus, as shown in Figure 8.

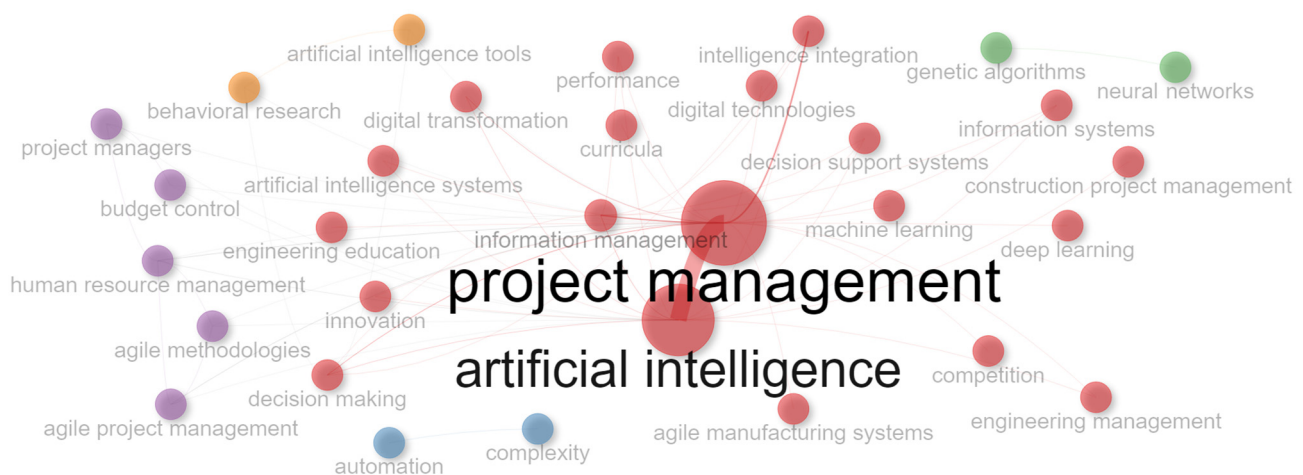


Figure 8. Keyword co-occurrence network.

The red cluster represents the central area of the network, highlighting terms such as “artificial intelligence”, “project management”, “machine learning”, and “digital transformation”. This cluster emphasizes AI’s transformative role in digitizing workflows, enhancing decision-making processes, and driving innovation within projects. The presence of keywords like “deep learning” and “decision support systems” represents the increasing use of cutting-edge machine learning methods to maximize performance and accomplish project goals. Organizations can use these technologies to forecast trends, evaluate complex data, and make data-driven decisions that support their strategic objectives.

The purple cluster focuses on “agile methodologies”, “human resource management”, and “budget control”, demonstrating how AI may be used to improve resource allocation, ensure adaptation in changing circumstances, and refine iterative project agendas. AI gives project managers the ability to make well-informed modifications through the use of real-time monitoring and predictive analytics, which lowers risks and improves financial control. Furthermore, the focus on human resource management draws attention to AI’s potential to simplify workforce allocation, boost team output, and guarantee effective task distribution throughout project lifecycles.

The orange cluster delves into the intersection of AI and human behavior, with keywords like “artificial intelligence” and “behavioral research”. This cluster highlights how crucial it is to create AI systems that complement behavioral and cognitive processes in order to improve teamwork and decision making. These AI-powered solutions are becoming more and more user-focused, which facilitates smooth workflow integration and improves interpersonal relationships in project teams. This cluster demonstrates the potential of AI as a computational tool and a collaborator in human-centric project management by identifying team issues and fostering collaboration.

The green cluster is centered around advanced computational techniques, featuring terms such as “genetic algorithms” and “neural networks”. This cluster investigates how AI may improve scheduling, predictive analytics, and resource allocation in intricate project environments. These advanced methods enable project managers to tackle complex issues including cost reduction, task balancing, and project outcome predictions in the face of uncertainty. In particular, neural networks allow the system to learn from past data, continuously enhancing and perfecting methods for project planning in the future.

Lastly, the blue cluster, which includes “automation” and “complexity”, underlines AI’s role in simplifying intricate processes, reducing manual efforts, and improving efficiency in highly interconnected and dynamic projects.

4.2. Trend Analysis

The trend topics analysis reveals the temporal evolution of key themes, offering insights into how the focus areas within the field have shifted over time (Figure 9). Here, several trends can be stated.



Figure 9. Trend topics.

On the one hand, the terms “digital transformation” and “artificial intelligence” gained significant traction during 2022–2023, highlighting their role in reshaping traditional PM practices through automation and efficiency. Subsequently, “information management” and “decision making” became critical themes in 2023–2024, emphasizing the need for robust data-driven frameworks. Large-scale data generation by businesses needs efficient data management for better project outcomes, resource optimization, and informed decision making. This change denotes an evolution toward evidence-based methods, where strategic choices in contemporary PM techniques are guided by AI and advanced analytics [44].

On the other hand, “agile project management” has also seen a steady rise, aligning with industry demands for flexibility and iterative processes [45]. The simultaneous rise of

“information management” underscores the necessity of robust data-handling capabilities to support agile practices, further demonstrating AI’s role in enabling such capabilities. Agile methodologies, in conjunction with AI, have also been used for digital transformation initiatives, highlighting the dynamic integration of technology and iterative workflows [39]. The continued focus on “decision making” is significant because it emphasizes how crucial it is for AI to support operational and strategic decisions. AI-powered predictive modeling has been extremely helpful in anticipating possible project delays, resource shortages, and cost overruns. Moreover, the expanding trend in “information management” suggests a growing recognition of data as a strategic asset, with AI enabling real-time monitoring and intelligent insights [46,47].

4.3. Thematic Map and Clusters

Figure 10 presents a clustering of documents based on their coupling, showcasing three distinct groups that are color-coded for clarity. The pink cluster focuses on terms like “model”, “cost”, and “big data”, suggesting a group of documents emphasizing the use of models for cost analysis and data management in the context of project management. The high confidence levels (100% for “model” and “cost” and 75% for “big data”) indicate strong coupling within this cluster. These elements are likely central to discussions on predictive analytics and financial estimations using large datasets in PM. The blue cluster contains “artificial intelligence”, “analytic hierarchy process”, and “systems” with a perfect 100% confidence in their coupling, except for “systems” at 33.3%. This suggests that this cluster is dedicated to exploring the integration of artificial intelligence in decision-making frameworks, particularly through structured methods like the analytic hierarchy process and systems thinking. Finally, the green cluster merges “project management”, “artificial intelligence”, and “information management” with 100% confidence across all terms, emphasizing the role of AI in enhancing project management practices through better information handling and decision-making capabilities. This cluster reflects the growing trend of AI applications directly improving efficiency and management strategies in project-related tasks.

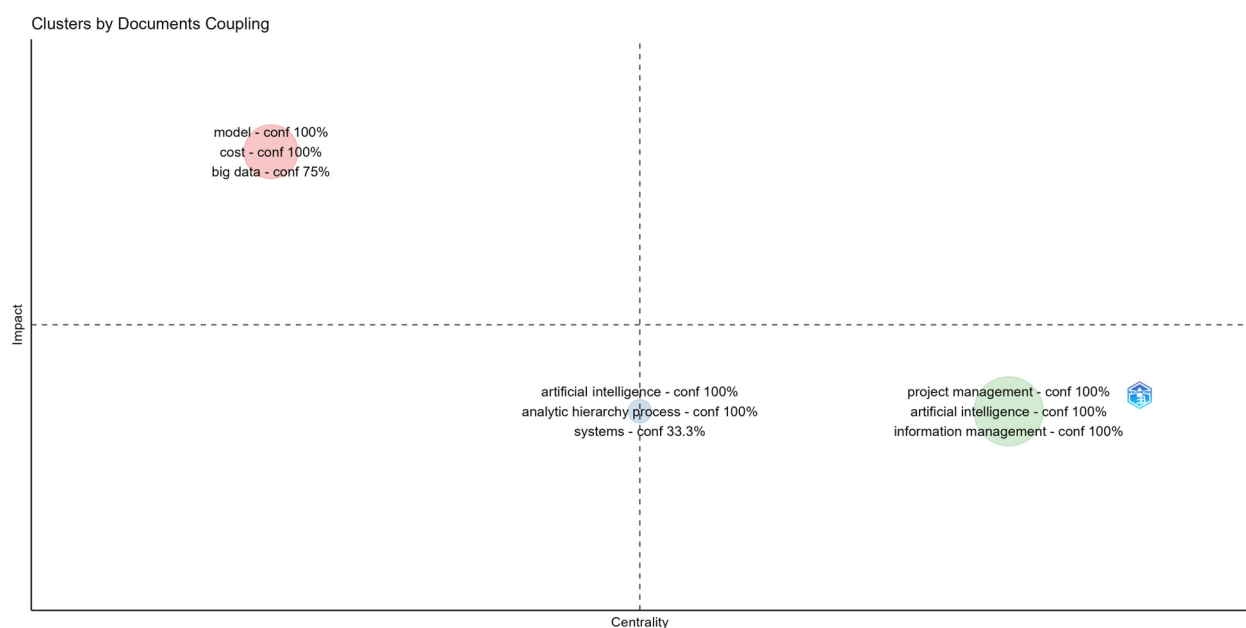


Figure 10. Document clusters.

The thematic map classifies various themes within the research field into four quadrants based on their development and centrality, as shown in Figure 11. The Niche Themes

in the upper-left quadrant, such as “information management”, “information systems”, and “management information systems”, are highly developed but less central to the broader field. These issues are specialized and may be particularly pertinent in specific situations, such as industries concentrating on digital transformation or systems management, but they do not presently dictate the overarching trajectory of the discipline [48].

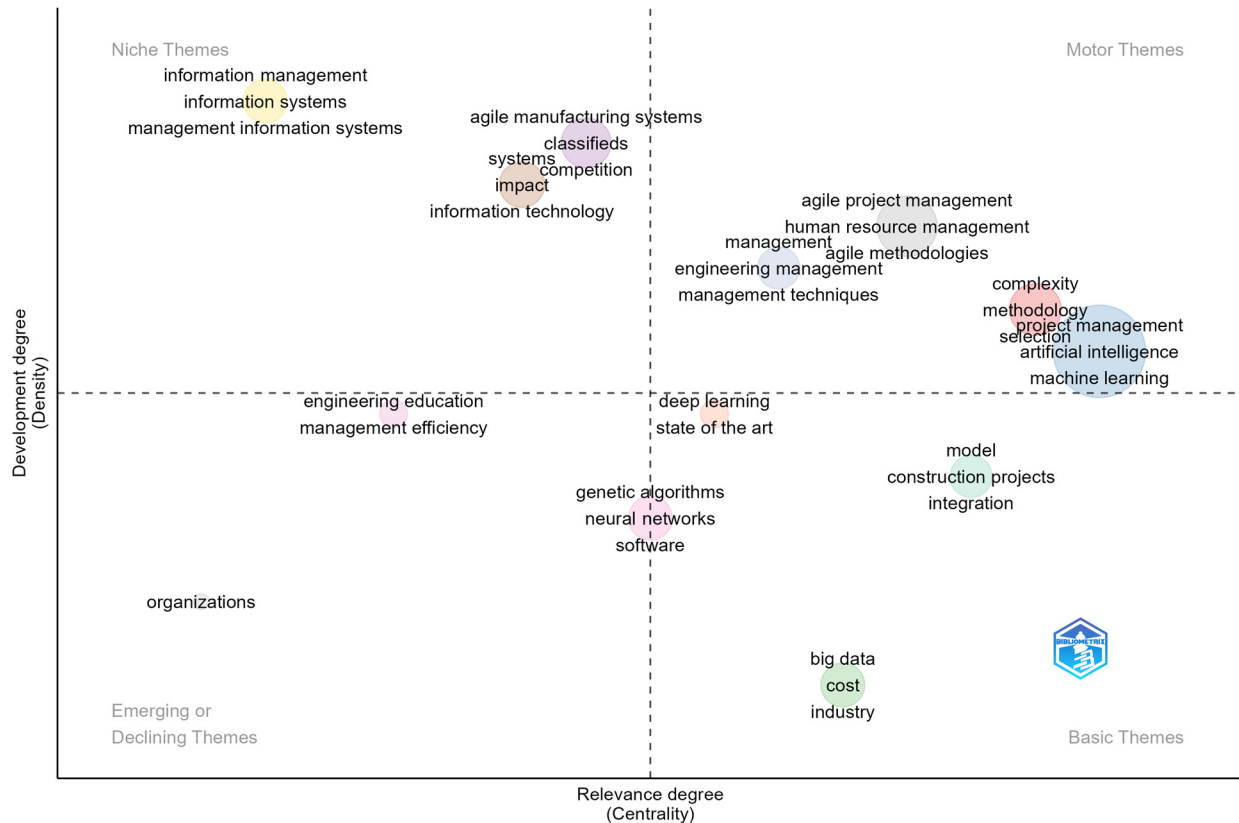


Figure 11. Thematic map.

In the upper-right quadrant, the Motor Themes are both well developed and central to the research field. Themes like “agile project management”, “human resource management”, and “artificial intelligence” fall into this category. These topics are crucial in forming the area and will probably drive new developments in the future. Their centrality demonstrates their impact on several fields of study, advancing project management through the incorporation of novel technology and approaches. For example, AI’s contributions to agile methodologies include predictive analytics for sprint planning, enhanced team collaboration through AI-driven tools, and continuous feedback mechanisms. Additionally, agile methodologies benefit from tools like Kanban boards integrated with AI to estimate lead times, optimize workflows, and improve overall efficiency [43].

The lower quadrants highlight themes that are either foundational or in flux. The Basic Themes in the lower-right quadrant, such as “deep learning”, “big data”, and “cost” are central to the field but still developing. They represent the foundational knowledge upon which the field builds, with potential for further exploration and application. Deep learning, for instance, has been explored as a means to facilitate “Construction 4.0” by enhancing automation, predictive capabilities, and project integration [49]. Big data allows comprehensive trend analysis and scenario simulations, offering organizations deeper insights into project environments. Conversely, the Emerging or Declining Themes in the lower-left quadrant, like “engineering education” and “organizations”, are less developed and central, indicating areas that might either be gaining interest or becoming less relevant.

4.4. Future Prospects and Challenges

AI plays a pivotal role in the enrichment of PM, which is characterized by high levels of complexity, with recent studies having highlighted its benefits [6,50,51]. Studies have also indicated the close relationship of AI with data science and big data [42,52,53] and its ability to optimize PM [54] and support agile PM [55]. The outcomes of this study further validate those of the literature which highlight the ability of AI to improve the accuracy and effectiveness of decision making, automation, risk, resource and time management, and monitoring, as well as of data collection, processing, and analysis [36,56,57]. Among the various AI subfields, particular emphasis is being placed on examining its related techniques, methods, and models [36,51]. Besides the focus on machine learning and deep learning, the research regarding the use of generative artificial intelligence in project management and the influence that it can have on the PM process and on project managers is increasing [5,58]. However, given the nature of the data examined in this study, there is a need to further explore how different machine learning and deep learning frameworks influence AI-driven PM initiatives.

The implementation of AI in project management presents significant technical, ethical, and practical challenges. A key concern is algorithmic bias, which can lead to unfair decision making, particularly in resource allocation and risk assessment [59]. For instance, AI models may prefer project kinds or teams due to past biases in training data, which would reinforce systemic inequities. Additionally, AI-driven automation raises concerns about workforce displacement, as routine project management tasks such as scheduling and reporting become increasingly automated [60]. While AI enhances efficiency, over-reliance on automated decision making can reduce human oversight, leading to ethical dilemmas when AI forecasts suggest project delays or cost overruns. A case study in the construction industry highlights this issue, where AI-based risk assessment models recommended significant project slowdowns, leading to conflicts between automated efficiency and human decision-making priorities [61]. Furthermore, while data privacy concerns are often mentioned, a more in-depth examination is necessary, particularly regarding how sensitive project information is stored, shared, and protected on AI-driven platforms. Addressing these challenges requires a balanced approach that integrates AI capabilities with human expertise, ensuring ethical, transparent, and effective project management practices.

Although this study emphasizes the importance of AI in project management, it has some limitations, including the difficulty of keeping up with the quick development of AI technologies. Future research should examine how AI might be used to improve sustainability in project management, specifically in terms of maximizing resource use, reducing waste, and promoting energy-efficient procedures. Furthermore, real-time resource tracking may be made possible by combining AI and IoT, which would enhance decision-making, efficiency, and transparency during a project. For project management procedures to be fair and transparent, ethical issues like bias reduction and responsible AI use should also be a top priority. Additionally, encouraging more international cooperation will help create AI-driven frameworks that are more flexible and inclusive, addressing regional issues and integrating different viewpoints to improve global project management techniques.

5. Conclusions

This bibliometric analysis of AI in PM from 2019 to 2024 reveals significant trends and understandings into this rapidly evolving field. With a 70.32% annual increase in publications, the research area had impressive growth. Given its increasing significance in handling the complexity of contemporary projects, this trend highlights the growing interest in incorporating AI technologies to improve PM methods.

The research is dominated by three major issues in literature: information management, decision-making, and machine learning. These fields emphasize the use of cutting-edge computational methods and show how they can enhance project management approaches' accuracy, efficiency, and strategic planning.

Geographically, the United States, China, and India stood out as the top three countries in terms of publication output, demonstrating their strong involvement in the subject. Nonetheless, the nations with the highest citation rates were South Korea, Australia, Vietnam, and the United Kingdom, indicating their prominence and proficiency in this field. This combination of high impact and prolific output is indicative of a complementary but geographically diversified research endeavor.

Recent patterns show a move toward data-driven strategies with a heavy emphasis on information management and decision-making, especially in 2023 and 2024. These advancements demonstrate the growing dependence on AI to analyze large amounts of information and produce useful insights, which are essential for strategic alignment and efficient project execution. As new tools for project management, technological developments like data science and generative AI are attracting interest.

Notwithstanding these developments, there is still opportunity for progress in international cooperation since the international co-authorship rate is still comparatively low at 11.3%. Strengthening international partnerships could further enrich the research landscape and foster a more unified approach to solving global PM challenges.

In conclusion, there are many subject clusters within the research landscape, such as AI-enhanced decision-making frameworks, cost analysis and data management, and AI applications in project management methods. These clusters provide important insights into the different ways AI might be used to improve project management, reflecting the diversity and interconnection of the research activities. Moving forward, researchers should focus on integrating AI with sustainable project management practices, exploring its potential to minimize environmental impact and optimize energy use. Additionally, the combination of AI and the IoT presents significant opportunities for real-time resource tracking, improving transparency and operational efficiency. For practitioners, adopting AI-driven strategies can lead to more data-driven decision making, streamlined workflows, and enhanced project outcomes. With all factors considered, this investigation highlights how AI in project management is evolving dynamically.

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